Recommendation System Using Amazon Reviews 2023 Dataset

1. Introduction

In this project, a recommendation system was developed using the Amazon Reviews 2023 Video Games dataset.

The project includes:

- Exploratory Data Analysis (EDA)
- Content-Based Filtering
- Collaborative Filtering
- Hybrid Recommendation System
- Feature Selection
- Model Evaluation using Cross Validation

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2. Dataset

The Amazon Reviews 2023 Video Games dataset was used.

The original dataset is very large, so a sampling strategy was applied.

Sampling Strategy

Load first 200k lines of the dataset.
Keep users who have at least 5 reviews.
Keep products that have at least 5 reviews.
Merge review data and metadata into a single dataset.

The sampled dataset was uploaded to Hugging Face and loaded directly from Hugging Face during execution.

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3. Exploratory Data Analysis

3.1 Dataset Overview:

The following information was examined:

```
Number of unique users: 8386
Number of unique items: 3762
Number of ratings: 52710
```

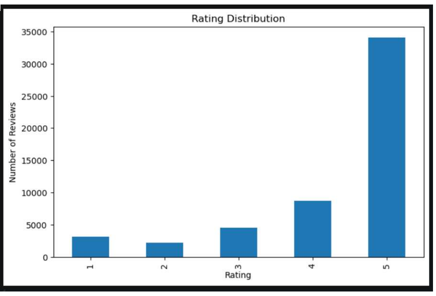
Review data fully complete:

```
Missing values in sampled reviews:
rating          0
title           0
text            0
images          0
```

Ratings:

Mean is very high which might be a drawback.

```
count    52710.000000
mean         4.294840
std          1.164437
min          1.000000
25%          4.000000
50%          5.000000
75%          5.000000
max          5.000000
Name: rating, dtype: float64
```



User Analysis:

Besides some extreme game lovers, mean and std values look perfectly normal.

```
User interaction statistics:
count    8386.000000
mean        6.285476
std         6.324014
min         1.000000
25%         3.000000
50%         5.000000
75%         7.000000
max       128.000000
Name: count, dtype: float64
```

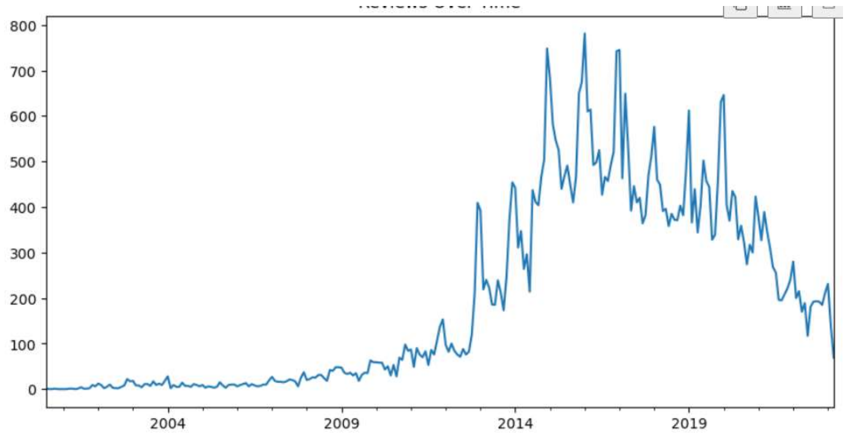
Product Analysis:

Popular items received significantly more reviews.

```
Item interaction statistics
count    3762.000000
mean       14.011164
std        17.746475
min         5.000000
25%         6.000000
50%         8.000000
75%        15.000000
max       296.000000
```

Temporal Analysis:

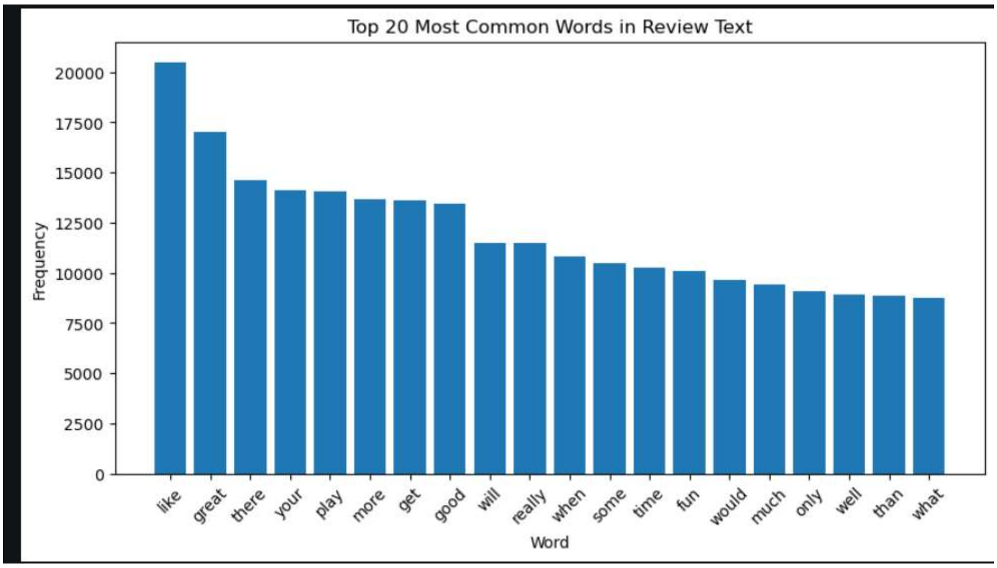
User review patterns show fluctuations possibly with the release of new games, peaks were reached in 2015s period then gradually declined.



3.6 Text Analysis:

Review texts and product descriptions were analyzed using TF-IDF representation.

Below are the most used words in the reviews:



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4. Recommendation Models

4.1 Content-Based Filtering:

Content-based recommendations were generated using product descriptions and textual information.

TF-IDF was applied to product text.

Cosine similarity was used to calculate similarity between products.

Products with similar textual features were recommended to users.

4.2 Collaborative Filtering:

Collaborative filtering was implemented using user-item interaction data.

The model recommends products based on the behavior of similar users.

4.3 Hybrid Recommendation System:

A hybrid approach was developed by combining content-based and collaborative filtering scores.

The hybrid model attempts to benefit from the strengths of both methods.

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5. Feature Selection

Below algorithms were used in selecting features:

Chi- Square

Information Gain

Relief

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6. Evaluation

6.1 Cross Validation:

Content based, Collobarative filtering and Hybrid filtering algorithms are designed

Firstly MAE and RMSE were applied (runtime 2m 23s)

Recommender algorithms were then redesigned to include Precision K and Recall K metrics (runtime 107 minute)

5-Fold Cross Validation was applied in all algorithms.

6.2 Performance Metrics:

The following metrics were used:

- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- Precision@K
- Recall@K

MAE and RMSE measure prediction accuracy.

Precision@K and Recall@K evaluate recommendation quality.

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7. Results

The performance results of Content-Based, Collaborative Filtering, and Hybrid models were compared.

	Model	MAE	RMSE	Precision@K	Recall@K
0	Content-Based Filtering	0.713298	1.120614	0.009851	0.058210
1	Hybrid Recommendation	0.748985	1.110818	0.011684	0.060923
2	Item-Based Collaborative Filtering	0.784232	1.182228	0.009164	0.046702

8. Discussion

Observations:

- Content based has the best MAE, therefore predicting closest to ratings in absolute error. This means the item information like metadata and texts alone seem to create a good enough recommendation algorithm.

- Hybrid recommender has the lowest RMSE. Since RMSE penalizes errors more quadratically the hybrid model may be better at escaping very wrong prediction mistakes.

- Content based may miss products behaviorally related, two items maybe bought at the same time even though having completely different descriptions, whereas collobarative filtering can recognize such patterns. Howeber since the matrix is sparse, item- item similarity also becomes very weak. Therefore collobarative did not predict anything better than the other two.

- Hybrid Recommender has the highest Precision K and Recall K. The Hybrid models improve the drawbacks stated above by using both signals therefore it has the valuable metrices like these the highest, as we observe %20 improvement.

Conclusion:

- Content based may have the highest MAE but the other 3 metrices are the best at hybrid recommender algorithm. Thus we can safely claim Hybrid algorithm performs best in this scenario.